exocomp

DESIGN CANVAS · V0.1 · IMPLEMENTATION REFERENCE

One audited queue. Nothing runs outside it.

Every artboard of the exocomp UI canvas, one page each. Values are authoritative in `internal/web/tokens.css` (shipped as text alongside this PDF) — this document carries layout and intent, not numbers to re-type.

01 Foundations

02 Accent study

03 Dashboard · dark (native)

04 Dashboard · light

05 First-day dashboard

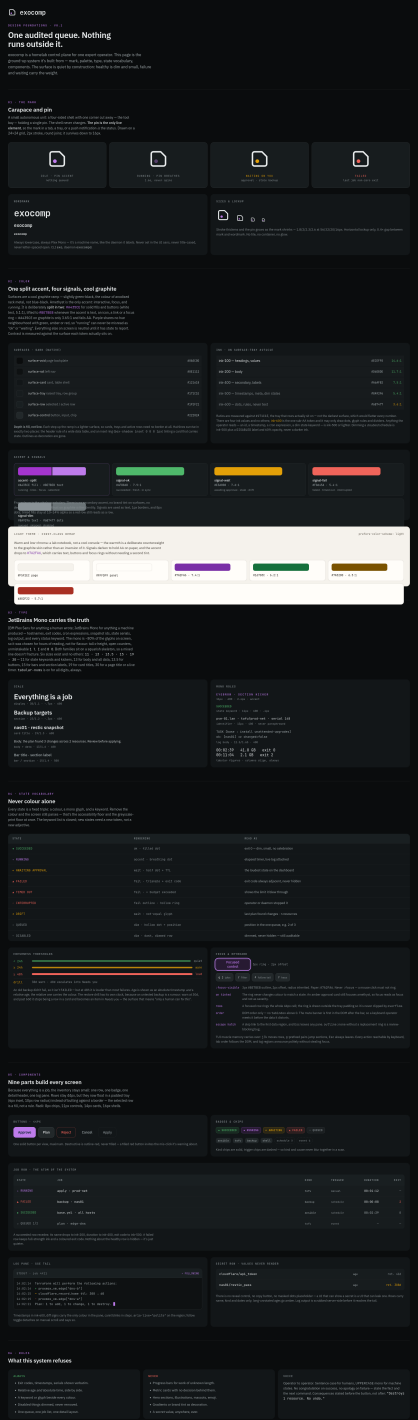
06 Job detail

07 Events

08 Phone approval set

09 Setup wizard · step 1

10 Recovery kit (print)


exocomp
One audited queue. Nothing runs outside it.
Carapace and pin
One split accent, four signals, cool graphite
JetBrains Mono carries the truth
Everything is a job
Backup targets
nas01 · restic snapshot
Never colour alone
Nine parts build every screen
What this system refuses

TURN 2 · TYPE

The mono is the decision

On our screens the mono is roughly 60% of the glyphs — every hostname, timestamp, exit code, cron expression and log line. The sans only sets headings and a sentence or two of prose. So if one family changes, it should be that one. Comfort is a mono comes from a tall x-height, open apertures and slightly wide spacing: the eye stops decoding and starts reading. Plex Mono is a beautiful typewriter face, but it's narrow and low x-height, which is exactly why our log tails feel technical rather than easy.

Below: the same three elements — a title, a table row, a log tail — set three ways at production sizes. Everything else is identical, and none of these is Geist, so we stay clear of the reference system's type as well as its purple.

2c Plex Sans + Plex Mono

apply · tofu/prod-net

The plan found 3 changes across 2 resources. Review before applying.

▸ RUNNING apply · prod-net 01:12

14:02:14 Plan: 1 to add, 1 to change, 1 to destroy.
14:02:15 process_vk_pem: edge["dev-b"]: Still creating...
14:02:09 Creation complete after 31s [id=8199]
00:02:39 41.8 GB exit 0 s148

Narrow, low x-height, typewriter DNA. Excellent tabular figures and very institutional — it reads as infrastructure, which is on-brief but not comfortable over long log sessions.

2b Plex Sans + JetBrains Mono even-only swap

apply · tofu/prod-net

The plan found 3 changes across 2 resources. Review before applying.

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14:02:14 Plan: 1 to add, 1 to change, 1 to destroy.
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00:02:39 41.8 GB exit 0 s148

Driven for reading code for hours: about 8% taller x-height at the same size, rounder counters, slashed zero, clearly distinct 0, 1, I and O. The big pairs gets noticeably calmer at zero cost to density — being monospaced, its figures are tabular by definition.

3c Monospe + JetBrains Mono both swapped

apply · tofu/prod-net

The plan found 3 changes across 2 resources. Review before applying.

▸ RUNNING apply · prod-net 01:12

14:02:14 Plan: 1 to add, 1 to change, 1 to destroy.
14:02:15 process_vk_pem: edge["dev-b"]: Still creating...
14:02:09 Creation complete after 31s [id=8199]
00:02:39 41.8 GB exit 0 s148

Monospe is the soft, high-x-height geometric that's sort of "comfy" products use it to translate their files — and slightly less serious. Its roundness fights the mono a little, where Plex Sans and JetBrains Mono share a square skeleton.

Recommendation

2b — swap the mono, keep the sans. It buys most of the comfort (the big tell and every table cell get easier) while keeping Plex Sans's slightly institutional tone for the prose, which suits a tool that reports rather than sells. Two families still, both open-licence, and the pairing shares a squarer skeleton so mixed lines don't fracture. 3c if you want the whole surface friendlier. 3a if the typewriter register is part of what you liked.

One consequence worth knowing: JetBrains Mono runs slightly wider per character than Plex Mono, so the fixed-table columns need about 4–6% more width. I'll hold that into the same pass as the 2b comfort changes rather than doing two rounds of column maths.

TURN 3 · WHY THAT APP FEELS CORPY

Comfort is four mechanisms, not more whitespace

What reads as "comfy" in that interface is mostly the absence of things: almost no visible borders, very few type sizes, no wide-tracked capitals, and nothing touching an edge. Depth comes from lightness steps instead of hairlines, radii are one step larger and consistent, and the whole screen uses about three text sizes in one or two greys. Density is not the variable — their list rows are about as tight as ours.

Three of those four are free for us. The fourth — softening state colour and lowering contrast further — we should refuse, because our brief pays for loudness on failure. Below: the same dashboard fragment as built (2a) and with the comfort pass applied (2b). Identical content, identical row heights, identical facts.

2a As built 11 type sizes · hairlines everywhere

NEEDS YOU · 2

▸ AWAITING APPROVAL expires in 10:41

apply · tofu/prod-net

Destroys 1 resource. No undo.

Review plan Reject

THE QUEUE

STATE	JOB	ELAPSED
▸ RUNNING	apply · prod-net	01:12
▲ FAILED	backup · nas01	exit 2
◦ QUEUED 1/1	docker-hosts.yml · nas01	–

- Sizes in scope: 10, 10.5, 11, 12, 12.5, 13, 13.5, 14 — eight in one card.
- Four hairline borders plus a 3px row track, all competing for the same job.
- Two wide-tracked mono capitals used as section furniture, not as data.

2b Comfort pass 4 type sizes · depth by fill

Needs you: 2

▸ AWAITING expires in 10:41

apply · tofu/prod-net

Destroys 1 resource. No undo.

Review plan Reject

In the queue 2 active · 1 waiting

▸ RUNNING	apply · prod-net	01:12
▲ FAILED	backup · nas01	exit 2
◦ QUEUED 1/1	docker-hosts.yml · n…	–

- **Four sizes:** 11 (data), 13 (everything), 13.5 (buttons), 17 (title).
- **No hairlines.** Surfaces step #0A0C0D → #131618 → #171C1E → #191F21; the amber edge is an inset ring, so it reads as lit rather than outline.
- **Section labels in sentence case sans.** UPPERCASE mono is now reserved for states — which makes it mean something again.
- **Rows float in a padded tray** (4px inset, 10px row radius) instead of butting against a border, and the active row is a fill, not a rule.
- **Radii up one step** — 6 chip | 11 control | 14 card | 16 shell. Same 44px rows, same facts, same exit codes.

What I would not take from it

Their calm comes partly from being almost monochrome — one white accent, no semantic colour, soft low-contrast text. We can't buy that: an old backup has to shout at 4th, and a failed row has to out-weigh nine healthy ones. So the comfort pass stops at structure and type; signal colour, contrast ratios and the state vocabulary stay exactly as specified. The other thing to leave behind is glow and blur — atmospheric on a landing page, dishonest on a status surface.

If 2b is right, the is a foundation-level change, not a screen tweak: it collapses the type scale, replaces the hairline system with a four-step surface ramp, and re-scopes uppercase-mono. Say go and I'll roll it through Foundations and the Web UI before building the remaining surfaces, so the new screens are born in the calmer system rather than converted later.

TURN 5 · ACCENT & WARM LIGHT STUDY

Three accents against a warmer paper

Same fragment six times: the approval card, three queue rows, a link and a focused control — once on warm paper, once on graphite. The neutral ramp is identical in all three; only the accent moves. Warm paper reads as a lab notebook rather than a cool console, and it does not clash with the dark rail in so long as the paper stays low-chroma and the accent has its own darker paper value. Contrast is annotated on every pairing.

paper: #F5F1E8 page · #FFFBF4 panel · #FAF7F1 row · #EFEBE3 hov | ink: #1D1A16 · #3A3328 · #6B6558 | signal: #1E7E9C · #7A1200 · #4A2F12

1 Cyan · as built #1B6F8D / #0A3A4A

Coolest of the three and the only one that sits next to the green "ok" hue — fine for a single operator, slightly less separable for deuteranopes. Dark text on the solid button.

2a Amethyst · your #A435CE #8E7BE8 / #7A2FA5

Solid fills and buttons stay #A435CE with white text (5.1:1). Accent-as-text links lift to #8E7BE8 — #A435CE on graphite is only 3.6:1 and fails AA. Purple sits far from all four signal hues, so nothing competes.

3 Dusty violet · calmest #636ACF / #4E54C5

Same hue family as 1b at roughly half the chroma — the "less bright" end. Reads as instrument-grey with a violet cast; the running dot is noticeably quieter than a failure, which suits a surface meant to be dull most of the time.

WARM PAPER

▸ AWAITING expires in 10:41

apply · tofu/prod-net

Destroys 1 resource. No undo.

Review plan Reject

▸ RUNNING	apply · prod-net	01:12
▲ FAILED	backup · nas01	exit 2
▪ SUCCEEDED	base.yml · all hosts	02:39

plan 0c31 ↗ focus ring 5.3:1 · AA

GRAPHITE

▸ AWAITING expires in 10:41

apply · tofu/prod-net

Destroys 1 resource. No undo.

Review plan Reject

▸ RUNNING	apply · prod-net	01:12
▲ FAILED	backup · nas01	exit 2
▪ SUCCEEDED	base.yml · all hosts	02:39

plan 0c31 ↗ focus ring 13.9:1 · AAA

WHERE I'D LAND

1b — purple is the better functional choice here, not just the one you asked for: it's the only accent that shares no hue neighbourhood with green, amber or red, so "running" can never be misread as "ok" or "waiting". Warm paper suits it better than it suits cyan, too. Keep #A435CE as the solid-fill and button colour with white text, and use the lifted #8E7BE8 wherever the accent is text, an icon, a link or a focus ring. If 1b still feels hot on a large dark dashboard, 1c is the same decision two steps quieter.

One caveat worth stating plainly: #A435CE is the reference system's brand purple. Using it as our accent puts exocomp visually adjacent to it — the mark, wordmark, warm-paper skin and graphite ramp are ours, but the accent should not be. Say the word and I'll shift the hue ~15° toward a more magenta or more indigo violet so it reads related-but-distinct.

exocomp

Operate

- Dashboard
- Jobs 1
- Approvals 1

Configure

- Workspaces ≠2
- Schedules
- Backups ▲1
- Secrets
- Events
- Recovery kit

● exocompd 0.7.2

pve-01.lan · up 6d 04h
dead-man · pinged 12m ago
ntfy · reachable 2m ago
data dir 71% · 38G free
queue 2 · oldest queued 14m

Update 0.7.4 ↗
snapshot first

Status pve-01.lan · homelab

● 1 RUNNING 14:03 UTC+11 Resume reactive Run...

○ REACTIVE PAUSED · GLOBAL ◐ + 1 SCOPED SILENCE no expiry set

Two overlapping mutes. The global pause has **no expiry** — set one, or nothing reactive will ever run again. The scoped silence on `pve-02 · storage` lifts in 2h 14m. **11 alerts swallowed since 11:46.**

Resume reactive | Edit silence

Needs you · 2

◐ AWAITING APPROVAL expires in 17:41, then auto-rejects

Review plan | Reject

apply · tofu/prod-net

Gated by rule `tofu-apply-requires-human`. Triggered by alert `dns_records_drifted` from Alertmanager at 13:58:04.

+1 add ~1 change -1 destroy plan 8c31 · serial 148

▲ BACKUP STALE 61h nas01 · restic · last ok 19 Aug 01:15

Three scheduled runs failed with `exit 2` — repository locked by a stale process. Nothing has been written to the NAS target since Tuesday.

Open last job | ▲ Force unlock...

Force unlock breaks the repository lock and runs as its own audited job. If the holding process is alive, the snapshot it is writing can be corrupted.

In the queue · 1 running, oldest queued 14m — all jobs →

STATE	JOB	KIND	TRIGGER	ELAPSED	EXIT
● RUNNING	apply · prod-net	tofu	manual	00:02:31	–
○ QUEUED 1/1	docker-hosts.yml · nas01,pve-02	ansible	schedule ↗	–	–

Backup freshness 3 legs · 24h / 48h · drill 30d, needs you at 60d

nas01 · restic	▲ FAILED	61h
b2 · exocomp-offsite	● OK	7h
git · state+config	◐ STALE	29h
restore drill · nas…	◐ UNTESTED	34d

Workspaces last plan

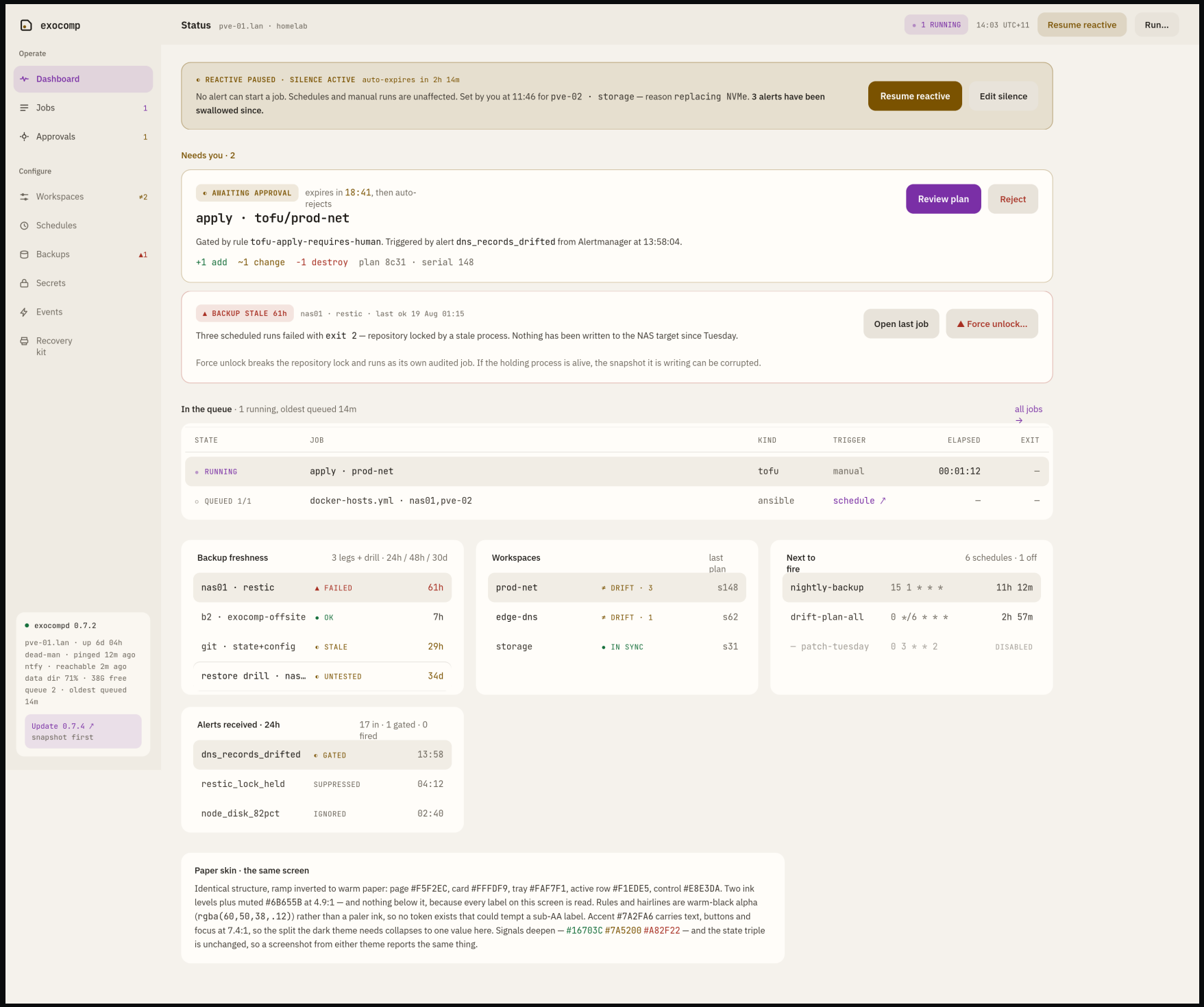
prod-net	≠ DRIFT · 3	s148
edge-dns	≠ DRIFT · 1	s62
storage	● IN SYNC	s31

Next to fire 6 schedules · 1 off

nightly-backup	15 1 * * *	11h 12m
drift-plan-all	0 */6 * * *	2h 57m
– patch-tuesday	0 3 * * 2	DISABLED

Alerts received · 24h 17 in · 1 gated · 0 fired

dns_records_drifted	◐ GATED	13:58
restic_lock_held	SUPPRESSED	04:12
node_disk_82pct	IGNORED	02:40



Paper skin · the same screen

Identical structure, ramp inverted to warm paper: page #F5F2EC, card #FFFDF9, tray #FAF7F1, active row #F1EDE5, control #E8E3DA. Two ink levels plus muted #6B655B at 4.9:1 — and nothing below it, because every label on this screen is read. Rules and hairlines are warm-black alpha (rgba(60,50,38,.12)) rather than a paler ink, so no token exists that could tempt a sub-AA label. Accent #7A2FA6 carries text, buttons and focus at 7.4:1, so the split the dark theme needs collapses to one value here. Signals deepen — #16703C #7A5200 #A82F22 — and the state triple is unchanged, so a screenshot from either theme reports the same thing.

exocomp

Operate

- Dashboard
- Jobs
- Approvals

Configure

- Workspaces
- Schedules
- Backups
- Secrets
- Events
- Recovery kit

exocompd 0.7.4
pve-01.lan · up 2h 14m
dead-man · pinged 4m ago
ntfy · reachable 1m ago
data dir 1% · 118G free
queue 0 · idle

Status pve-01.lan · installed 2h ago

○ QUEUE IDLE 14:03 UTC+11 Pause reactive Run...

Needs you · 0

- Nothing is waiting on a human. When an approval, a stale backup, or an overdue drill needs you, it lands here first.

In the queue · idle all jobs →

STATE	JOB	KIND	TRIGGER	ELAPSED	EXIT

Nothing queued. Every run — manual, scheduled, or reactive — lands in this one queue first.

`exocomp run <playbook>` — your first job will appear here

Backup freshness 1 leg · thresholds 24h / 48h

nas01 · restic	• OK	2h
restore drill · nas…	• PASSED	2h

One leg is one copy — `exocomp backup add-target` when you have offsite storage.

Workspaces 0

— NONE YET

No workspaces yet — `exocomp workspace add` when you have OpenTofu to manage.

Next to fire 0 schedules

— NONE YET

Today's backup ran from setup. `exocomp schedule add nightly-backup '15 1 * * *'` keeps it honest.

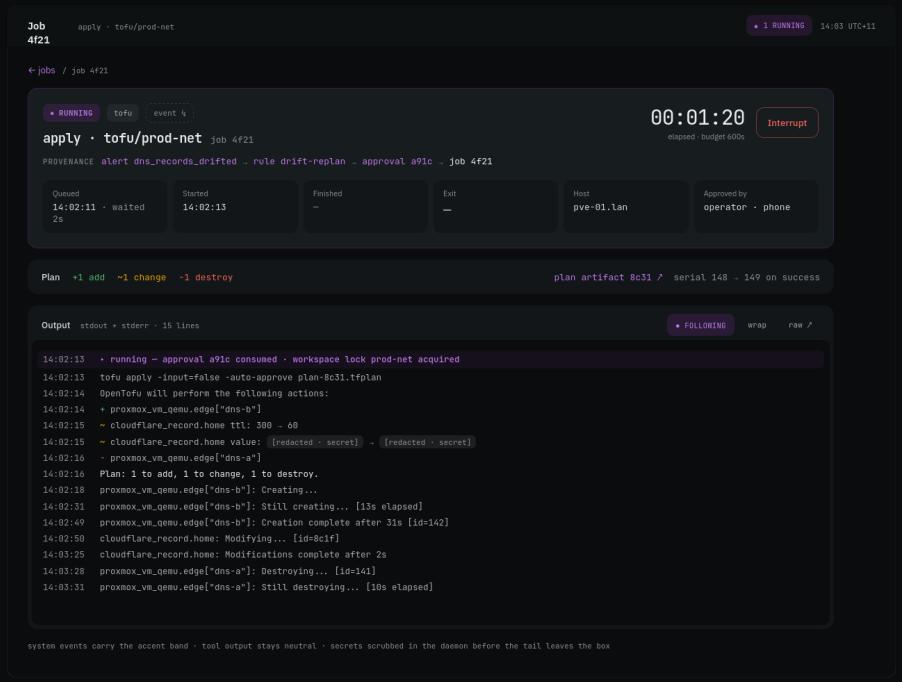
Alerts received · 24h 0 in · 0 fired

— NOTHING WIRED YET

Point Alertmanager at POST `/v1/events` and every alert lands here, fired or not.

day one · empty is a state, not an error — each card says what fills it, once

08 · JOB DETAIL — DARK · RUNNING · LIVE TAIL



08 · JOB DETAIL — LIGHT · FAILED · STDERR UP TOP



Events every alert received · what became of it

14:03 UTC+11

17 in 11 ignored 2 matched 3 suppressed 1 gated 0 fired

last 24h · nothing ran without being on this page first

all · 8 | ignored · 3 | matched · 1 | suppressed · 2 | gated · 1 | fired · 1

j k move · ↵ expand · e explain

RECEIVED	ALERT	SOURCE	MATCHED RULE	OUTCOME	REASON
13:58:04	dns_records_drifted	alertmgr	drift-replan	◐ GATED	awaiting approval a91c · TTL 30m

NORMALIZED PAYLOAD

```
{
  "source": "alertmanager",
  "name": "dns_records_drifted",
  "labels": {
    "zone": "lan",
    "severity": "warning"
  },
  "startsAt": "2026-08-22T13:58:01Z",
  "fingerprint": "c0ffee12"
}
```

EXPLAIN · RULE EVALUATION TRACE — evaluated in 3.1ms

```
rule drift-replan
  when  alert.name == "dns_records_drifted"        → true
  and   alert.labels.zone in ["lan","home.arpa"]   → true · "lan"
  and   !silences.match(alert)                     → true · none in scope
  cooldown 6h · last fired Wed 18:12               → eligible
  action plan(tofu/prod-net)                       → plan 8c31 · +1 ~1 -1
  gate  tofu-apply-requires-human                  → approval a91c · TTL 30m
outcome GATED · nothing applies without a human
```

12:40:11	node_disk_82pct	alertmgr	—	IGNORED	no rule matched
11:52:30	smart_pending_sectors	alertmgr	disk-health-observe	● MATCHED	log only — rule has no action
11:47:02	zfs_pool_degraded	alertmgr	storage-remediate	◐ SUPPRESSED	swallowed · silence pve-02 · lifts 16:00
04:12:57	restic_lock_held	healthchk	backup-unlock-retry	◐ SUPPRESSED	cooldown 6h · next eligible 14:22
02:40:18	node_disk_82pct	alertmgr	—	IGNORED	no rule matched
Wed 23:58	deadman_ping_missing	healthchk	deadman-restart	● FIRED	job 4e88 · succeeded · exit 0
Wed 22:03	watchdog	alertmgr	—	IGNORED	no rule matched · alertmanager heartbeat

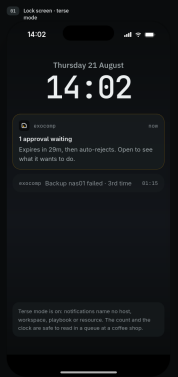
showing latest 8 of 231 events · retained 90d · the enum is closed: ignored / matched / suppressed / gated / fired

PHONE POSTURE · ONE DECISION, ONE THUMB

Notification → context → decision

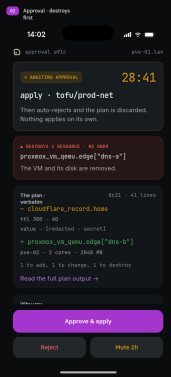
Seven screens, ten seconds on the happy path. The phone is not a small dashboard — it shows exactly one thing that cannot proceed without a human, with enough of the real plan to judge it. Nothing here is a summary of a summary: the destroy comes first, the diff is the daemon's own output, and every target is ≤52px. Terse mode assumes the operator is in public: the lock screen names no host, no workspace, no runbook.

01 Lock screen · terse mode



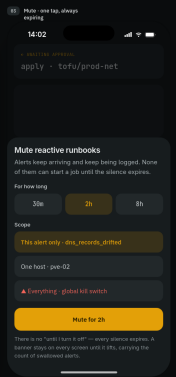
Two notifications, ranked by whether a human is the blocker. The approval carries its own countdown and the fail-closed consequence; the failure below it is information, not a demand. Tapping either deep-links straight to screen 02 — no inbox, no list.

02 Approval · destroys first



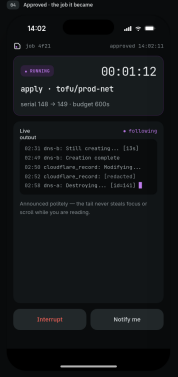
Order of the screen is the order of the risk: state and clock, then the destroy in its own red block, then the rest of the diff, then why it's asking. Secret-backed values render as [redacted · secret] — the phone never carries a value this well. Approve is 56px and alone; Reject and Mute sit below at 52px so a thumb can't confuse them.

03 Mute · one tap, always expiring



The dangerous option is reachable but ruined honestly — a global kill is a separate, red row, not a wider version of the first one, and it never fires from this tap: it opens the confirm at 07. Duration is mandatory, so the product has no state where a human has silenced it forever and forgotten.

04 Approved · the job it became



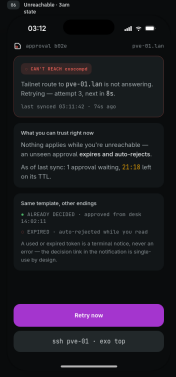
Approving does not end the interaction — it hands over a job id and the same tail the desk gets, so the decision can be watched to its end state. Notify" as exists because the honest answer to "will this take long?" is often yes; the operator should be able to put the phone away.

05 Stale plan · apply refused



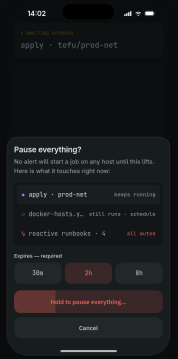
The one safety state the happy path hides: state moved between plan and tap, so the countdown is gone — there is nothing left to approve, and no disabled Approve button pretending otherwise. The screen says what moved, who moved it, and offers the only two honest actions: re-plan against serial 149, or discard.

06 Unreachable · 3am state



At 3am this state is the product. It leads with what is safe to believe — fail-closed means an unreachable daemon can't hurt you — then the freshest cached facts with their age, then the fallback an expert actually uses: the exact SSH command, copyable. The terminal notices (already decided, expired) reuse this template with a different keyword.

07 Global kill · second confirm



"Everything" never fires from one tap. The confirm shows what the kill actually touches — and what it doesn't: the running apply finishes, schedules still fire, only reactive runbooks mute. Expiry is required, and commit is hold-to-confirm (the bar fills over 1.5s), which a pocket cannot do by accident. The pause itself lands as an audited job.

exocomp setup · pve-01.lan

Quit setup — nothing written yet

1 Identity
where the key lives

2 Recovery kit
◐ requires the printed page

3 First backup
run one, restore one

4 Hand over
the daemon takes the wheel

SETUP · STEP 1 OF 4

Where should this machine keep its key?

One key locks everything exocomp stores — job history, configuration, the secrets your runbooks use. Pick where it lives. Everything else is automatic.

◉ **On this machine** RECOMMENDED • TPM 2.0 DETECTED

A key file on pve-01, sealed to this machine's own security chip — unlocked automatically at boot, useless on any other hardware.

▸ Why would I change this?

◐ BEFORE STEP 2 CAN CONTINUE

Next, exocomp prints a one-page recovery kit. **The wizard will not go past step 2 until you confirm that page is on paper.** If this machine dies, that page is the only way back in.

Changeable later: `exo identity migrate` — one command, no data loss.

Continue → print the recovery kit

exocomp · RECOVERY KIT

pve-01.lan · kit v3 · generated 2026-08-22 · replaces v2 (destroy it)

FIRE SAFE ONLY.
Verify quarterly:
exo kit verify

This page regenerates every secret exocomp manages. With it — and any one backup target below — the whole system can be rebuilt on new hardware. Commands are verbatim: type them exactly as printed.

1 · RECOVERY KEY (SECRET — THIS IS THE ROOT)

```
AGE-SECRET-KEY-
1QGFZ RCV8A NW3J9
LUXKD
4TP2M H7SEW 06QYV
WNZ5F
8GJXR E2DAK L9M4U
CQ
```

Spaces are cosmetic — strip them when typing. QR encodes the same key.

2 · RECIPIENTS (PUBLIC — FOR VERIFICATION)

kit v3 · this page

```
age1u9dx zmr4wnl2 c5qsy0e8 3htfv
ag6 7kpj9042 sr8dwc3l n5xqme7s
```

operator laptop

```
age1qdw7 nc44hjkl m3pv9820 srtxy
e56 gzfau0ml edk93c2h 8n5vqsr7
```

3 · SALT (PUBLIC — NEEDED FOR EVERY DERIVATION)

```
9f3a 71c2 0b5e d688 42aa 17f0
c39d e51b
7c04 88d2 fa61 30e9 55b7 c8f3
1d26 a04e
```

kit sha256 · 3b1f 09e4 77aa 41d0 · **exo kit verify** checks this page against the daemon

4 · READ-ONLY STORAGE CREDENTIAL (WRITE BY HAND)

5 · DERIVATION RECIPE (GENERAL — ONE RULE)

```
secret = HKDF-SHA256(
  key  = recovery key (1),
  salt = salt (3),
  info = "exocomp/v1/" +
purpose )
```

```
# restic repository password
exo kit derive restic/nas01
# ansible-vault password, any
vault-id
exo kit derive ansible-vault/
homelab
# every purpose in use, from
state
exo kit derive --list
```

Works offline: **exo kit derive** asks for sections 1 and 3, nothing else.

6 · TARGETS (WHERE THE DATA IS)

```
rest:https://nas01.lan:8000/exo
comp
b2:exocomp-offsite:/nas01
ssh://git@github.com/tom/homela
b-state.git
```

7 · DEAD-MAN PING (SILENCE > 15M PAGES YOU)